

AI DECLARATION FORM

For this lab, I used Claude (an AI assistant by Anthropic) as a learning guide to help me better understand concepts and navigate the lab instructions.

HOW I USED IT

The lab instructions were clear and straightforward to follow. I was able to work through most of the setup and exercises by carefully reading and following the provided guidelines. However, there were moments where I wanted a deeper understanding of why certain things worked the way they did.

For example, in Exercise 4 when implementing the normal equation for linear regression, I did not just want to know the formula but wanted to understand why matrix multiplication is used in that context and what it means conceptually when we compute theta using $(X^T X)^{-1} X^T y$, and other concepts. The AI broke down each part of the equation and explained what role each matrix operation played in finding the best fit coefficients.

In Exercise 5, when filling missing values using group by, I asked why filling by group mean is a better approach than filling with the overall dataset mean. The AI explained how using the group mean preserves the natural variation between different majors rather than introducing bias from the overall population.

When I encountered errors I shared them with the AI and asked for an explanation of the root cause rather than just a fix, so I could understand what went wrong and avoid repeating it.